

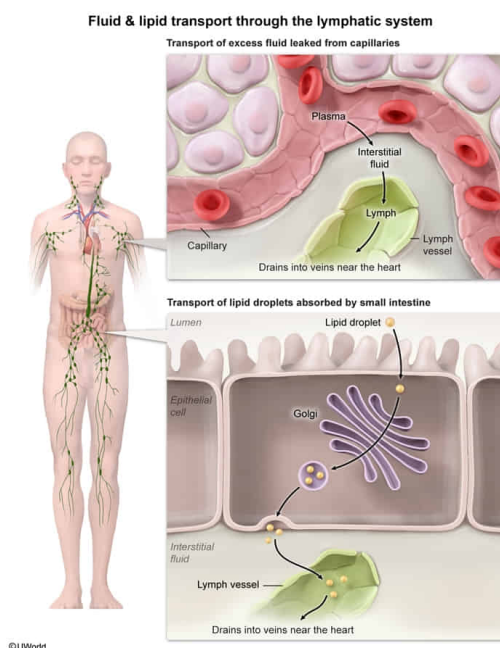
Following digestion in the small intestine, certain lipids are absorbed by intestinal epithelial cells and packaged into large droplets. These lipid droplets are released from intestinal epithelial cells and transported into the lymphatic system to ultimately drain into a large vein near the heart. Which series shows the order in which these lipid droplets are transported from the intestine to the bloodstream?

- ☐ A. Lymph capillaries → interstitial fluid → lymph vessels → lymph duct → vein [7%]
- ☐ B. Interstitial fluid → lymph vessels → lymph duct → lymph capillaries → vein [23%]
- ✓ ☒ C. Interstitial fluid → lymph capillaries → lymph vessels → lymph duct → vein [61%]
- ☐ D. Lymph capillaries → interstitial fluid → lymph duct → lymph vessels → vein [6%]

Correct

61% answered correctly

Explanation:



When blood flows through capillaries, some fluid containing plasma proteins leaks out of the blood vessels and into the interstitial fluid. The interstitial fluid fills the space between blood vessels and surrounding cells. Leaked excess fluid from the capillaries, now also known as interstitial fluid, must be returned to the bloodstream to maintain the proper blood volume and protein concentration. To reenter the bloodstream, the fluid must first pass from the interstitial space and enter a network of vessels and nodes known as the **lymphatic system**. Once in the lymphatic system, this fluid is now known as **lymph** and is ultimately drained into two large veins near the heart.

In addition, the lymphatic system also collects large lipid droplets absorbed by the small intestine. Certain lipids **digested** in the small intestine are absorbed by intestinal epithelial cells and packaged into large droplets. However, their large size prevents them from crossing directly into the capillaries surrounding intestinal cells. Instead, these lipid droplets are transported through the lymphatic system and into the bloodstream as follows:

1. Lipid droplets are released from the epithelial cells into the **interstitial fluid**.
2. **Lymph capillaries** collect lipid droplets from the interstitial fluid.
3. Lymph containing the lipid droplets then flows from the capillaries into increasingly larger **lymph vessels**.

4. Lymph is then transported into a large tubule structure called a **lymph duct**, which drains into a **large vein near the heart**. As a result, the lipid droplets within the lymph enter the bloodstream and circulate throughout the body (**Choices A, B, and D**).

Educational objective:

The lymphatic system collects protein-containing fluid leaked from blood capillaries and transports it back to the bloodstream. In addition, large lipid droplets absorbed by the small intestine are also transported to the bloodstream via lymph vessels. Fluid is collected by lymph capillaries, flows into larger lymph vessels, and is transported into lymph ducts that drain into veins near the heart.

Time spent:0 sec

Subject:
Biology

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System:
Circulation and Respiration